



SIMATS Online Programming Lab

DS PROGRAMMING



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QUESTIONS

10 / 10 completed

Bin. Search-Iterative-Temp

BIN. SEARCH-ITERATIVE-TEMP

"Read n daily temperatures into an array, sort ascending using any method. Use binary search (iterative) to find first day $\geq 30^{\circ}\text{C}$ and last day $\leq 25^{\circ}\text{C}$. Display temperatures, indices (0-based), or ""Not found"". Validate sorted; $n \leq 50$.

Sample Input: 6

22 28 31 24 29 32

Sample Output (after sort):

22 24 28 29 31 32

First ≥ 30 : index 4 (31)

Last ≤ 25 : index 1 (24)"

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 #define MAXN 50
4
5 void binary_insert_sort(int a[], int n) {
6
7     for (int i = 1; i < n; i++) {
8         int key = a[i];
9         int j = i - 1;
10        while (j >= 0 && a[j] > key) {
11            a[j + 1] = a[j];
12            j--;
13        }
14        a[j + 1] = key;
15    }
16 }
17
18 int is_sorted_asc(const int a[], int n) {
19     for (int i = 1; i < n; i++) {
20         if (a[i] < a[i - 1]) return 0;
21     }
```

OUTPUT

```
22 24 28 29 31 32
First >=30: index 4 (31)
Last <=25: index 1 (24)
```

INPUT

6

22 28 31 24 29 32

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